Weisshorn nose cone at 720 mm: joint loads and temperatures

2027_C_FD_NOSE-CONE-720MM_LOADS-THERMAL — Fabio Feig (FD), 30.10.2026

These are the loads at the nose joint and the temperatures of the 720 mm nose (geometry sheet 2027_C_FD_NOSE-CONE-720MM_GEOMETRY), inputs for the ST design. They come from the FD Python models of 2027_C_FD_REPORT-2 (3-DoF ascent, semi-empirical drag, engineering heating). Two trajectories: *nominal* (ISA day, I_{sp} masses) and *hot day* (30 °C ground, LVO masses, the envelope). **Limit** loads are the flight values; **ultimate** = limit × FoS 2, the ERT convention (e.g. Firehorn nose load case 2024_C_SE_ST_NOSECONE_REQ_10: deceleration × dry mass × FoS 2). Requirements: Weisshorn sheet of 28.10.2026.

1 Inputs

Design	Profile [mm]	Payload [kg]	Nose mass m_n [kg]	Arm ℓ CP–joint [mm]
A	LV-Haack 679.0 + 41	4.34	5.74	425
B	Von Kármán 700.0 + 20	0.35	1.75	373

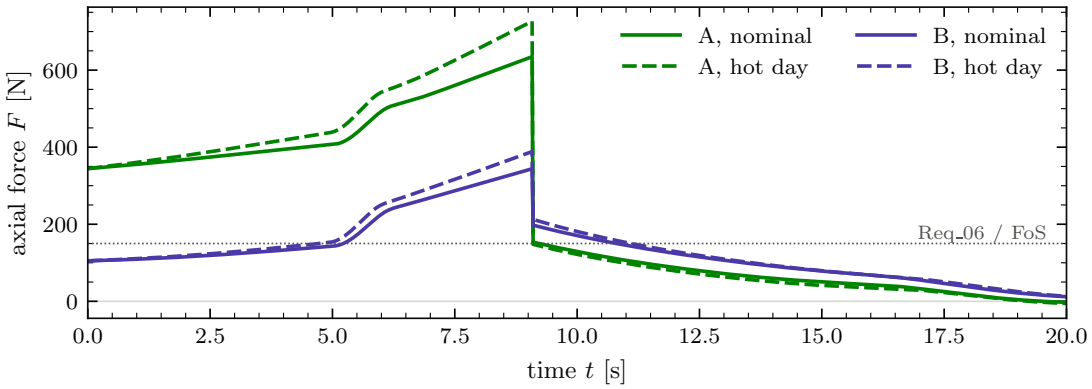
m_n = nose with payload structure at the Req_04 limit (1.40 kg) + payload carried in the nose (A: 3U GRV 3.99 kg + CanSat 0.35 kg; B: CanSat). $A_{\text{ref}} = \pi R^2 = 254.5\,\text{cm}^2$. ℓ = distance from the nose centre of pressure (slender body) to the joint plane.

2 Axial force at the joint

Free body of the nose along the axis (compression > 0), with T the thrust, D the vehicle drag, m the vehicle mass and D_n the nose drag:

$$F(t) = m_n \frac{T - D}{m} + D_n, \qquad F_{\text{rec}} = 20\,g\,m_n \times \text{FoS} \quad \text{(Req_07, recovery deceleration, tension).}$$

		Compression [N]					Tension, recovery [N]	
Case		lift-off	max (limit)	ult.	Req_06 300	min F	ult.	Req_07 2000
A	nominal	344	635 (9.1 s)	1270	**exceeds**	-4	2252	**exceeds**
A	hot day	344	727 (9.1 s)	1454	**exceeds**	-7	2252	**exceeds**
B	nominal	105	344 (9.1 s)	688	**exceeds**	0	686	ok
B	hot day	105	389 (9.1 s)	777	**exceeds**	0	686	ok



The maximum is at burnout (highest acceleration and drag); the nose stays in compression during the ascent ($F_{\min} \approx 0$), so the only tension case is the recovery load.

3 Bending moment and shear at the joint

Normal force of the nose at incidence α (slender body, $C_{N\alpha} = 2/\text{rad}$) at the maximum dynamic pressure, applied at the nose CP:

$$V = C_{N\alpha}\,\alpha\,q_{\max}\,A_{\text{ref}}, \qquad M_b = V\,\ell.$$

				Limit, per degree		M_b ultimate [N m]			
Case		$q_{\max}$ [kPa]	M	M_b [N m/°]	V [N/°]	$\alpha = 1°$	2°	5°	α at Req_12 [°]
A	nominal	116.8	1.47	44.1	104	88	176	441	2.3
A	hot day	131.4	1.56	49.6	117	99	198	496	2.0
B	nominal	117.6	1.47	38.9	104	78	156	389	2.6
B	hot day	132.5	1.56	43.9	118	88	175	439	2.3

Req_12 (200 N m ultimate) is reached at $\alpha = 2.0°$ for design A on the hot day: the incidence that ST designs for (gusts, weathercocking) decides whether Req_12 holds.

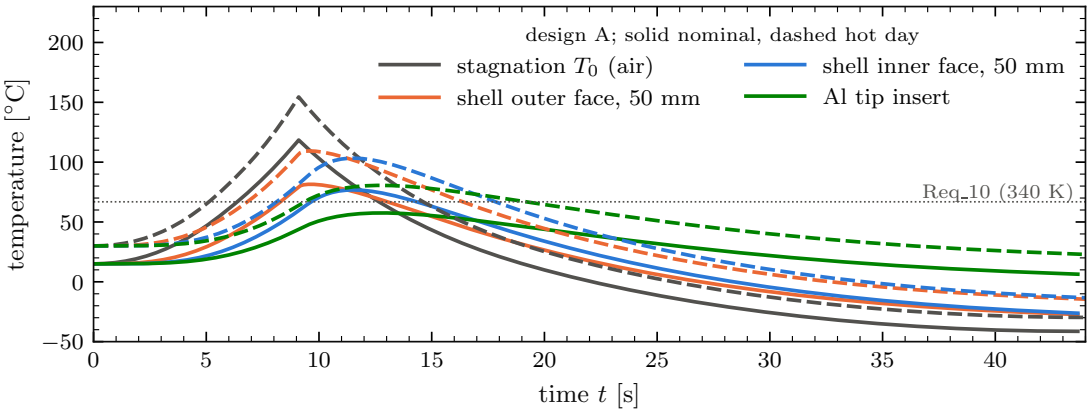
4 Wall pressure

The highest wall pressure is at the stagnation point of the tip, $p_0 - p_\infty$ behind the normal shock (max. at burnout), an upper bound for the whole shell: A 177 / 204 kPa; B 179 / 206 kPa (nominal / hot day). The pressure distribution on the shell, $C_p(x)$, follows from the Fluent runs of REPORT-5.

5 Temperatures

Laminar stagnation-point heating (Fay–Riddell) on the $r_n = 4.5$ mm tip, turbulent (Reynolds analogy, Eckert reference temperature) downstream; the Al 6082 tip insert (80 mm, 60 g) as a lumped mass, the 1.2 mm CFRP shell as 1D conduction with an adiabatic inner face (conservative, bare shell without the insert). Peak values over the flight [°C]:

	A		B			
Location	nominal	hot day	nominal	hot day	Limit	Worst case
Stagnation temperature T_0 (air)	118	154	119	156	–	–
Al 6082 tip insert (bond to the shell)	58	81	53	75	120 (DP490)	ok
CFRP shell outer face, $x = 50$ mm	82	109	82	110	–	–
CFRP shell inner face, $x = 50$ mm	77	103	77	104	67 (Req_10)	**exceeds**
CFRP shell inner face, mid-length	71	97	72	97	67 (Req_10)	**exceeds**
CFRP shell inner face, shoulder	69	94	69	95	67 (Req_10)	**exceeds**



Materials are the REPORT-1 assumptions (Al 6082 insert, CFRP shell 1.2 mm, DP490 bond): to be confirmed by ST. Fluent heat-transfer runs and an Ansys transient-thermal model of the tip will replace these values in 2027_C_FD_REPORT-5 (12.11.2026).

6 Open points for ST and SE

- **Req_06:** the ultimate compression exceeds 300 N for both designs (A 1454 N, B 777 N, hot day). **Req_07:** with the 3U GRV in the nose, the recovery tension of A (2252 N) exceeds 2000 N.
- **Req_12:** the design incidence at max q must be agreed (Req_12 is reached at $\approx 2°$).
- **Req_10:** does 340 K apply to the air in the nose or to the inner wall? The bare inner wall of A reaches 69–77 °C (nominal) against 67 °C.
- **Req_08:** the TBD values (max temperature, duration) can be filled from section 5.